

Ruiwen WANG

Research Engineer — ML Systems & AI Infrastructure

Distributed LLM training · Automatic parallelization

Contact

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Highlights

16,384 accelerators in production
up to 1.7× faster training
dense · MoE · multimodal models,
up to 1T+ parameters
7 publications · 3 patent families

Core Areas

Training: large-scale LLM pretraining · MoE · long context
Parallelism: DP / TP / PP / EP · ZeRO-style sharding · pipeline scheduling · recomputation
Planning: cost models (compute, memory, communication) · ILP optimisation · graph (IR) level
Performance: profiling · bottleneck analysis · tuning

Technical Skills

Languages: Python, C++
Frameworks: PyTorch, Megatron-LM, MindSpore, MindFormers
Systems: Ascend CANN, HCCL (NCCL-style collectives)
Tools: Linux, Git, ILP solvers

Awards

Best Poster Award — Euro-Par 2025
Golden Prize — Huawei, 2024
Letter of Appreciation
China Telecom · 2025

Languages

English · French · Mandarin · Cantonese

Online

🌐 ruiwen.wang
🌐 linkedin.com/in/ruiwen-wang-ai
📄 Google Scholar

EXPERIENCE

Ph.D. Researcher (CIFRE) — LLM Training Systems

2023 – 2026

Huawei Paris Research Center · Sorbonne Université & EURECOM · Paris

- Built automatic-parallelization planners combining compute, memory and communication cost models with optimisation; integrated into MindSpore / MindFormers (open-source AI training stack; [SAPP-PPB](#)) and benchmarked against Megatron-LM.
- Enabled production use through configuration guidance, debuggable traces and regression benchmarks across model families and cluster scales.

Research Apprentice → Research Engineer — AI Infrastructure

2021 – 2023

Huawei Paris Research Center · Distributed & Parallel Computing Lab · Paris

- Profiled and tuned large-scale LLM training and inference on production Ascend 910 NPU clusters, identifying compute, memory and communication bottlenecks.

DEPLOYMENTS & IMPACT

- Scale:** Ascend 910 clusters up to 16,384 NPUs — dense, MoE and multimodal models up to 1T+ parameters (DeepSeek, Llama, Mistral / Mixtral, Qwen).
- Production training:** supported China Telecom's DeepSeek-MoE deployment (Letter of Appreciation) and TeleChat3-MoE training.
- Portability:** cost surfaces separate model structure from hardware rates, enabling plans to retarget across clusters and accelerator families without re-profiling.

PUBLICATIONS

First-author publications

- Concord** — NeurIPS 2026. Reuses valid layer costs after model edits; 2.7× median re-planning speedup over full search, with identical plans in 54 evaluated cases.
- PRISM** — SCA/HPC Asia 2026. Profiling-free symbolic memory models guide parallel planning; 4.2–8.0% mean absolute percentage error in peak-memory prediction.
- BMPipe** — IEEE CLUSTER 2025. ILP-based joint optimisation of pipeline layer placement, recomputation and chunking; 1.7× training throughput with interleaving versus non-interleaved Megatron-Even on Llama-175B at 16,384 NPUs.
- H²O** — Euro-Par 2025, Best Poster Award. Two-level search jointly optimising parallelism degrees, micro-batching and pipeline configuration; +37% throughput over D-Rec and +22% over expert tuning on DeepSeek-141B (128 NPUs).
- ManuMatic** — IFIP NPC 2025. Operator-sharding hints with automatic graph-wide completion; 2.24× training throughput over D-Rec on Mixtral-8×7B (8 NPUs).

Co-authored publications

- MLLM Pipeline Bubble Modeling — SCA/HPC Asia 2026 (poster).
- TeleChat3-MoE — arXiv 2025 (technical report).

PATENTS

- Execution-plan generation — WO 2025/020165 A1 (PCT), Huawei; published 2025.
- Large-model load balancing — Huawei; pending publication.
- Multidimensional parallelism configuration — Huawei; pending publication.

EDUCATION

Ph.D. Computer Science (CIFRE) — Sorbonne Université

2023 – 2026

Defended 24 July 2026 · Research with EURECOM and Huawei Paris.

M.Sc. Computer Science — Sorbonne Université

2019 – 2022

B.Sc. Computer Science — Université Paris-Saclay

2016 – 2019